

SEO A/B Testing Analysis

Landing Page Optimisation Experiment · CTR Improvement Study

PROJECT

Athys Kitchen Diary

PLATFORM

WordPress Blog

TOOL

Python / Google Colab

FOCUS

CTR Optimisation

01 Project Overview

This project presents an SEO A/B testing analysis conducted to evaluate the impact of landing page title optimisation on organic search performance for the blog Athys Kitchen Diary.

The experiment compares an original landing page title (Version A) against a restructured, intent-driven title (Version B) to measure the difference in Click-Through Rate (CTR) from Google Search results.

Search performance data including impressions and clicks was collected and structured as a time series. The analysis was performed using Python in Google Colab, applying statistical testing to validate whether the observed improvement in CTR was significant.

02 Business Background

Athys Kitchen Diary is a personal WordPress blog publishing informational and recipe-based articles targeted at organic search growth. The blog covers topics in food, cooking, and healthy eating, attracting readers through search engines.

As the blog grew, it became clear that certain landing pages were receiving strong impressions in Google Search but converting poorly into clicks. This pointed to a title optimisation opportunity – pages were being seen but not clicked, suggesting the titles were not compelling enough to match user intent.

This project was initiated to test whether restructuring the page title with stronger, intent-aligned language would result in measurable CTR improvement.

03 SEO Goals & Hypothesis

Primary Objective: To determine whether optimising a landing page title improves organic search engagement (CTR).

Key SEO Goals	Hypothesis Framework
Increase Click-Through Rate (CTR)	H0 (Null): No significant difference in CTR between Version A and Version B H1 (Alternative): Version B results in a statistically significant improvement in CTR
Improve search result attractiveness	
Validate performance changes using statistical methods	
Demonstrate SEO experimentation capability	

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Audit Methodology

A structured SEO A/B testing framework was followed:

1	Dataset Preparation	Search performance data containing daily impressions and clicks was collected and organised into a time-series structure for both versions, aligned with the format used in Google Search Console exports.
2	CTR Calculation	CTR was calculated using the formula: $CTR = \frac{Clicks}{Impressions}$. This normalised performance across varying traffic volumes for a fair comparison.
3	Segmentation Analysis	Data was segmented into Control Group (Version A – original title) and Variant Group (Version B – optimised title). Performance metrics were aggregated at group level.
4	Data Visualisation	Two approaches were used: a bar chart for average CTR comparison between versions, and a line chart for CTR trend over time – identifying both overall performance difference and consistency of improvements.
5	Statistical Testing	An independent sample t-test was conducted using Python's SciPy library to confirm statistical significance, ensuring observed differences reflected a real performance change rather than random variation.

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Tools Used

Google Search Console – impressions and clicks data export	Google Colab – Python execution environment
pandas – data processing and aggregation	matplotlib – data visualisation (bar chart and line chart)
scipy – statistical hypothesis testing (independent sample t-test)	WordPress – Athys Kitchen Diary blog platform

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Key Findings

6.1 CTR Performance Comparison

Metric	Version A — Control	Version B — Variant
CTR Level	Lower and more stable	Consistently higher values
Representation	Baseline performance	Improved search engagement
Stability	Stable across time period	Stable, sustained improvement

6.2 Statistical Result

Statistical Test Outcome (Independent Sample T-Test)
p-value obtained from the t-test was below 0.05
The performance difference is statistically significant
Strong evidence to reject the null hypothesis (H0)
Version B is confirmed to deliver a genuine CTR improvement over Version A

6.3 Trend Analysis Insights

Trend Findings
Version B maintained consistently higher CTR across the entire time period
No major volatility observed in performance
Improvement was stable rather than temporary

07 Recommendations

7.1 Title Optimisation Strategy
Use structured, descriptive, and intent-driven titles similar to Version B
Include user intent keywords
Use benefit-oriented phrasing
Ensure clarity of topic in every title
7.2 Content Optimisation Expansion
Extend A/B testing methodology to meta descriptions
Test featured snippet optimisation
Test blog headings (H1, H2 structure)
Test internal linking anchor text variations
7.3 SEO Experimentation Framework
Implement monthly title testing cycles

Build a CTR monitoring dashboard
Use data-driven content updates based on test results

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Priority Action Plan

Priority	Action Item	Impact
High	Replace low-performing titles with optimised Version B style	High CTR improvement
High	Track CTR using SEO analytics tools	Performance monitoring
Medium	Expand testing to multiple blog pages	Scalability
Medium	Implement structured SEO testing framework	Long-term growth
Low	Test meta description variations	Incremental gains

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Expected Results

Based on the analysis, the following outcomes are expected from full implementation:

20%–40% increase in CTR	Improved organic visibility in search results
Higher engagement from search users	Better alignment with search intent
Increased traffic efficiency without additional impressions	

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Conclusion

This project demonstrates how SEO performance can be systematically evaluated using A/B testing combined with statistical validation techniques applied to real blog search data.

By comparing two versions of a landing page title on Athys Kitchen Diary, the experiment produced clear, measurable results: Version B significantly outperformed Version A in CTR, confirming that structured, intent-driven title optimisation directly improves organic search engagement.

The methodology used here — data collection, CTR normalisation, visual trend analysis, and t-test validation — reflects standard SEO experimentation practices used in data-driven marketing environments and can be scaled to any content website or e-commerce platform.

Project Files & Supporting Documents

Click any item below to open it directly on GitHub:

Version A	Landing Page — Original Title (Control)	View on GitHub
Version B	Landing Page — Optimised Title (Variant)	View on GitHub
Dataset	Search Performance Data — Impressions & Clicks (CSV)	View on GitHub
Python Notebook	A/B Testing Analysis — Google Colab Notebook	View on GitHub